SWC1IC: Short-Range Wireless Communication – WS 2026
Lab Report 1

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Abstract

This lab unit deals with various interesting problems ... (give a short summary of the specific topics addressed). If you decide to write your report in German, make sure to change the `\documentclass` parameter from `language=english` to `language=german`. If you write in MS Word (and such), try to use a similar structure.

1.1 The first assignment

The goal of this assignment is to analyze the wireless performance of the nRF52840 SoC by measuring signal strength (RSSI) and packet loss over different distances, transmit powers, and modulation schemes.

- **Setup and Implementation:**
  - *Hardware:* Two Nordic nRF52840-DK boards connected to a PC via USB (one transmitter TX, one receiver RX).
  - *Software:* Zephyr `radio_test` application controlled via PuTTY over serial UART (115200 baud, 8N1).
  - *Channel Selection:* Using the nRF Connect RSSI Viewer, an interference-free frequency of 2440 MHz was selected, set as channel offset 40 (`start_channel 40`).
  - *Transmission:* TX sent bursts of 1000 packets with pattern `pattern_11110000` using `start_tx_modulated_carrier 1000`.
  - *Metrics:* RX logged the received packets N_{RX} and RSSI using `print_rx`. Packet loss was calculated as:

$$\text{Packet Loss } [\%] = \frac{1000 - N_{\mathrm{RX}}}{1000} \times 100 \tag{1.1}$$

- **Test Procedure:** Measurements were taken across 5 spatial distances (d_1 to d_5), starting close where reception is perfect (RSSI > -60 dBm, 0% loss) and ending at a distance where packet loss becomes clearly visible (RSSI ≈ -95 dBm).

Summary and comments

Here you should summarize your learning experience, personal discoveries and major difficulties (if any). Also state the approximate amount of time you spent for this assignment and any optional accomplishments. This non-numbered section is produced with the `\section*{..}` command.